

Energy-Efficient Online Continual Learning for Time Series Classification in Nanorobot-Based Smart Health

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Abstract—Nanorobots have been used in smart health to collect time series data such as electrocardiograms and electroencephalograms. Real-time classification of dynamic time series signals in nanorobots is a challenging task. Nanorobots in the nanoscale range require a classification algorithm with low computational complexity. First, the classification algorithm should be able to dynamically analyze time series signals and update itself to process the concept drift (CD). Second, the classification algorithm should have the ability to handle catastrophic forgetting (CF) and classify historical data. Most importantly, the classification algorithm should be energy-efficient to use less computing power and memory to classify signals in real-time on a smart nanorobot. To solve these challenges, we design an algorithm that can Prevent Concept Drift in Online continual Learning for time series classification (PCDOL). The prototype suppression item in PCDOL can reduce the impact caused by CD. It also solves the CF problem through the replay feature. The computation per second and the memory consumed by PCDOL are only 3.572 M and 1 KB, respectively. The experimental results show that PCDOL is better than several state-of-the-art methods for dealing with CD and CF in energy-efficient nanorobots.

Index Terms—Concept drift (CD), nanorobot, online continual learning, sensor time series classification, smart health.

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I. INTRODUCTION

NANOROBOTS present the possibility of very effective new instruments for the treatment of human diseases and the enhancement of human biological systems [1]. For example, the nanomedical heart tracker can accurately track heartbeats [2]. The physical characteristics of organs and tissues can be represented in various physiological signals, such as the electrocardiogram (ECG) [3] and photoplethysmographic (PPG) [4]. Early prevention of diseases can be achieved through the timely detection of physiological signals [5]. Nanorobots based on nanoelectrical systems can convert physiological signals into electrical signals that can be transmitted back to the physician for early disease diagnosis [6].

A sequence of data points recorded in chronological order is called a time series. It is a dynamic data stream. The distribution of data streams is likely to change over time, called Concept Drift (CD) [7]. Threshold-based model discarding (ThMD) is a common method to solve CD [8]. It usually determines whether to update the model based on its prediction accuracy and the threshold [9]. The setting of the accuracy drop threshold size is crucial, but ThMD is not applicable for the recurrence concept and blip drift because they have a probability of recurrence. Additionally, ThMD cannot address the issue of Catastrophic Forgetting (CF) to achieve Online Continual Learning (OCL).

The ability to preserve past knowledge and the rapid adaptation of new knowledge are the focus of continual learning. For OCL, using experience replay [10] to overcome CF takes up a lot of memory space. The intelligent algorithms deployed on nanorobots should be energy-efficient and power-saving [11]. In numerous data mining applications, the initial dataset has a very large range of features, typically with a high dimension [12]. The processing of such high-dimensional data needs additional calculation time and storage space. Therefore, it is difficult to deploy and run replay-based continual learning algorithms on nanorobots.

OCL aims to learn new classes from a stream of data without forgetting previously acquired information. However, OCL may encounter problems such as data imbalance and CD, which require addressing both CF and CD. Continual Prototype Evolution (CoPE) [13] considers both issues by utilizing experience replay for small memory buffer sizes. However, CoPE doesn't

achieve good results in dealing with CF because it only uses a memory buffer to replay the data.

To solve these problems, this work explores energy-saving OCL technology to dynamically classify sensor time series in nanorobots. We propose an algorithm that prevents CD during online continual learning, called PCDOL. The loss function of PCDOL contains three components: (1) The prototype suppression item prevents abrupt changes caused by CD during prototype updating by utilizing the prototype suppression factor. (2) The feature sparsity item saves space for subsequent data by sparing the feature vectors in the hidden space and reducing the number of saved features. And (3) the feature replay item achieves the effect of overcoming CF by saving the training parameters. PCDOL can accommodate sudden drift, reoccurring concepts, and blip drift. It reduces the impact of abnormal data through the suppression of prototypes. It also reduces CF, computational complexity, and processing speed in OCL. This is a more comprehensive consideration than existing approaches. Our contributions include the following:

- Improved OCL ability: We propose a sensor time series classification algorithm, PCDOL, which can prevent CD and CF based on a real-time prototype updating schema.
- Advanced mechanisms for solving CD and CF simultaneously: We create a prototype suppression item that can reduce the impact of abnormal data like CD; design a feature replay protection mechanism to preserve the training parameters while saving the old data through a small memory buffer to reduce CF; and set a feature sparsity item to save space for subsequent data.
- Energy savings for nanorobots: PCDOL is computationally efficient and space-saving. The computation per second and memory consumed during real-time classification are 3.572 M and 1 KB, respectively. Low time and space complexities make it energy-saving and appropriate for running on nanorobots.
- Performance evaluation based on comprehensive experiments: We compare PCDOL with a number of state-of-the-art OCL methods. The average classification accuracy of PCDOL is around 18.7% better than the baseline methods. We also present a case study of applying PCDOL to classify different types of physiological signals collected by nanorobots.

The rest of this article is divided into the following sections: Section II introduces the related work. Section III contains a detailed description of the PCDOL algorithm. Section IV contains experiments and results, and the conclusion is shown in Section V.

II. RELATED WORK

A. Techniques of Nanorobots in Smart Health

Embedded nanotechnology-enabled sensors in innovative wearable devices allow for continuous tracking of physiological data [20]. Using neural nanorobots, it is possible to connect the human brain to a computer cloud system, also known as the “Brain/Cloud interface.” Neural nanorobots communicate brain signals to external machinery so that they can carry out

TABLE I
DEFINITION OF SYMBOLS

Symbol	Meaning
X_{nn}^{cN}	the nn-th sample of class c_N
P^{c_1}	prototype of class c_1
N_C	the number of classes in the memory
H	training set
S	data stream
g^{TCN}	feature extractor
P	prototype set
M_B	memory replay buffer
M_P	store prototypes buffer
τ, a, b, β	hyperparameters
ε, γ	prevent hyperparameters with zero denominator

the required tasks [21]. Surya et al. [22] described robots for neurorehabilitation of temporomandibular disorders (TMD) that acquire relevant EEG and EMG data through electrodes. Based on EEG and EMG signals, the exoskeleton can cure jaw diseases.

B. Sensor Data Classification Methods for Overcoming Catastrophic Forgetting

The goal of OCL is to maintain high accuracy on a task while also adapting to new data and potentially learning new tasks over time. It has the advantages of flexibility, convenience, and efficiency compared to traditional methods [23]. In OCL, data is provided in the form of streams [24]. In PCDOL, we use input in the form of task increments, with each task containing one data stream. The new task’s data may cause the model to perform worse on the old task. A number of approaches have been proposed to solve this problem [25]. Continual Unsupervised Representation Learning [16] shows strong competitiveness by mixing generative replay to combat forgetting.

C. Sensor Data Classification Methods Considering Concept Drift

The potential impact of the new data on the probability distribution is unknown, so CD is unpredictable [26]. Many methods have been used to detect drift. Nishida et al. [17] use a statistical test of equal proportions to detect different kinds of CD. The ensemble of subset online sequential extreme learning machine proposed by Mirza et al. [18] included a long-term memory that stores information and a drift detector. CD was studied in incremental learning by Elwell et al. [19].

D. Comparison With Other Methods

Table II compares PCDOL with other incremental classification methods based on four criteria: its ability to solve CD, its ability to overcome CF, its use of sparsity, and its energy efficiency. These works [14], [15], [16] do not consider CD in time series, while this work [17] is ineffective against CF. In contrast to [18], [19], PCDOL addresses both CD and CF by combining experience replay and knowledge distillation. It also establishes a feature sparsity item to preserve space for subsequent data and uses prototype suppression and automatic prototype updating techniques to save computing and storage

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B. Feature Sparsity

We propose a loss function based on the features of the latent space. First, we extract the sample into a multi-channel feature (F) by a two-layer convolutional neural network (line 14), as $F = \text{Conv}(H)$. Then, we normalize the feature vectors of a particular class by (7), both to unify the vector size to speed up the processing and to prepare for the subsequent sparse loss.

$$f_{c_i}^* = \frac{f_{c_i}}{\max_{f_{c_i}} + \gamma}, \forall i \in \{1, 2, \dots, N\}, f_{c_i} \in F. \quad (7)$$

$$L_{fs} = \frac{1}{|f_{c_i}^* \in F|} \sum \frac{\exp(f_{c_i}^*)}{f_{c_i}^* + \gamma}, \forall i \in \{1, 2, \dots, N\}. \quad (8)$$

where γ is the hyperparameter that keeps the numerator from being zero, we take $\gamma = 1e-15$. f_{c_i} is the feature vector of the c_i class in F . The maximum value of any of the feature channels for the c_i class is $\max_{f_{c_i}}$.

Because the exponential function is proportional to the data, we use (8). Theoretically, any stretching function can be used, but the exponential loss function has excellent monotonicity and non-negativity properties. Since normalization specifies the features in the range $[0, 1]$, only a small number of feature vectors are retained by feature sparsity. The larger the $f_{c_i}^*$, the smaller the L_{fs} . Feature sparsity improves classification accuracy and reduces the amount of data stored in the latent space. We use the item to concentrate features on fewer channels, leaving space in the latent space for subsequent classes.

C. Feature Replay

We design replay only for the old task, which can improve the replay effect and cause no waste of memory. Therefore, when the first task is performed (line 17), the model does not perform replay operations. We choose to save the predictive label (O_{T-1}) trained on the old task ($T-1$). We use the O_{T-1} of the old task ($T-1$) to overwrite the predicted label O_T of the current task (T). Then, we get an update on new prediction labels (O'_T). The model has the ability to adapt to the new class without forgetting about the old class. The feature replay loss is expressed as (9) [27].

$$L_{fr} = - \sum_H \left[\sum_{t=1} P^{c_i} \log g^{c_i} + (1 - P^{c_i}) \log (1 - g^{c_i}) + \sum_t \beta_{y=P^{c_i}} \log g^{c_i} + (1 - \beta_{y=P^{c_i}}) \log (1 - g^{c_i}) \right]. \quad (9)$$

where β is the label parameter containing the processed task. The loss is divided into two parts: the old task and the current task. One is the cross-entropy of the predicted labels and inputs of the old task. The other is the cross-entropy of the predicted labels covering the parameters of the old task and the inputs of the new task. We begin the training process for the next task with the parameters completed from the previous task's training. This step makes sure that the learned parameters will not be forgotten during the continuous learning process.

IV. EXPERIMENTS

The experiment environment is: Python 3.9, Pytorch 1.10.1, and Windows 10. The hardware environment is: NVIDIA Geforce RTX2080 Ti GPU, Intel I9-9900 K CPU, 16 GB of Memory, and a 1 TB SSD hard disk.

A. Data Set

According to the Association for the Advancement of Medical Instrumentation (AAMI) [28], we classify the ECG data into ten categories. We use the following datasets: MIT-BIH Arrhythmia Database (AR) [29], MIT-BIH Supraventricular Arrhythmia Database (sup) [30], St Petersburg INCART 12-lead Arrhythmia Database (incart) [31] and MIT-BIH Long-Term ECG Database (long) [31] are common arrhythmia data sets of ECG. Wrist PPG During Exercise (PPG) [32] is PPG data.

The data we use can be collected by nanorobots. Gandhi et al. [34] and Mamdiwar et al. [35] described the use of nanoelectrodes and nanorobot technology to collect ECG and PPG data. By deploying our algorithms on nanorobots, real-time data analysis and classification can be achieved [36].

Our experiment design is as follows:

- Since the sampling rate varies across datasets, we unify the sampling rate at 128 Hz or 360 Hz. A wavelet transform is used to denoise the physiological signal data to reduce the impact of noise on model fitting.
- We artificially generate several datasets, which include three types of CD: sudden drift, reoccurring concepts, and blip drift [19], to verify the performance of PCDOL. Sudden Drift: The DS1 dataset consists of two categories, long and AR, without duplicates. The model initially learns the class of long and then uses the subsequent AR data, which has a different distribution, to simulate mutant data. This change in data distribution is not expected to occur again. The DS2 dataset is composed of incart and AR. Incart and long have different sampling rates than DS1. Incart and AR have the same category but different shapes. Reoccurring Concepts: The DS3 dataset includes AR and incart data sets. We insert the incart data into fixed positions in the AR data stream and revert back to the original data after each drift. The incart data from different distributions occurs cyclically. The DS4 dataset consists of AR and sup. The settings are the same as DS3. Blip Drift: In the DS5 dataset, we consider this type of drift abnormal data and use processed ECG data as an anomaly sample to be inserted into the original AR data.
- We validated its ability to support OCL using the real data sets AR and PPG.

B. Experimental Settings

To simulate the OCL process, our data stream consists of a series of tasks $T = \{T_0, T_1, \dots, T_n\}$, and OCL is achieved through task increment. We divided the physiological signal data into five tasks, each containing two class labels. Since our dataset itself is unbalanced, the largest N class contains 68,569 samples, while the J class contains only 75.

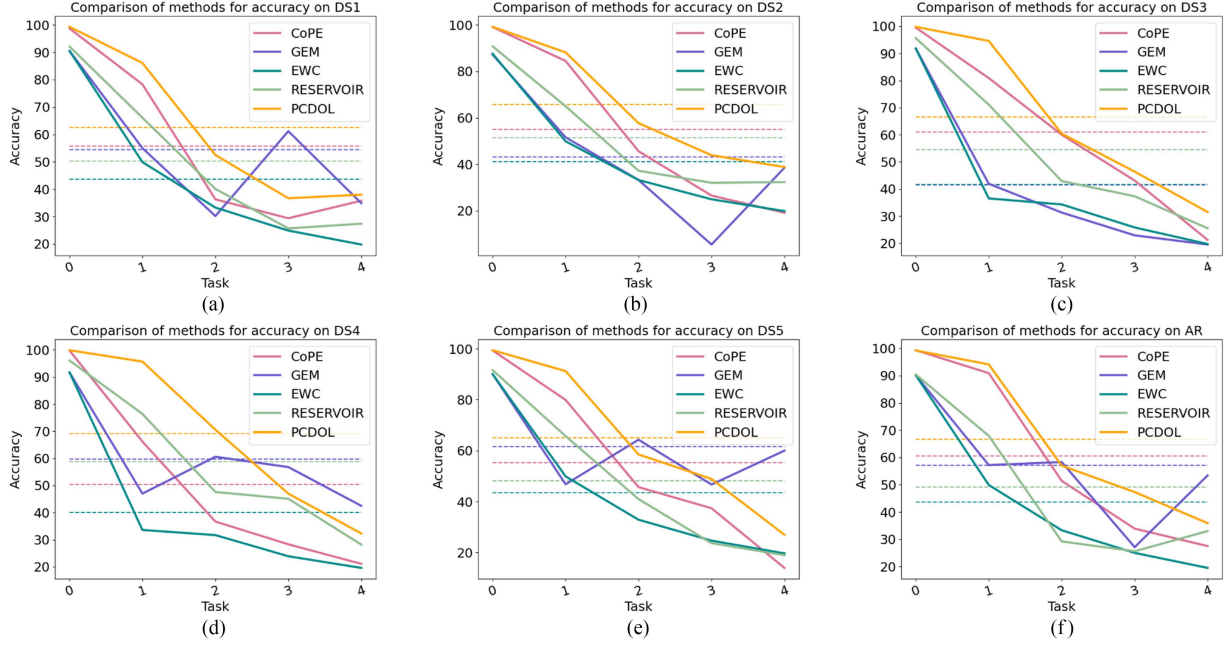


Fig. 2. Accuracy(%) for each task. (a) to (e) correspond to the accuracies based on DS1 to AR of different methods.

TABLE III
ABLATION EXPERIMENT

Method	accuracy(%)				
	task0	task1	task2	task3	task4
L_{fr}	81.37	51.57	35.66	18.67	18.12
L_{con}	99.61	82.81	51.58	41.98	48.37
L_{fs}	88.07	43.15	29.06	22.75	15.83
$L_{fr} + L_{con}$	95.07	75.69	46.22	36.09	46.20
$L_{fr} + L_{fs}$	90.13	45.01	30.04	22.56	19.82
$L_{con} + L_{fs}$	99.72	89.49	28.20	16.88	19.82
PCDOL	99.56	94.39	54.54	45.50	52.55

The bold values indicates the maximum values of the column.

We divided the ECG data stream into five tasks, with two categories within each task. The PPG data stream is divided into four tasks, each containing only one class of data.

We use the TCN network as a feature extractor and use the results of the output layer as the features of this sample. In the testing phase, the prototypes and parameters will no longer be updated.

C. Experimental Details

1) *Compare With Existing Work*: CoPE: considers the problems of CF and CD. GEM: applicable to unbalanced datasets. EWC: judges which parameter is more important for the old task. RESERVOIR: implemented for large data streams with lots of data or infinite data.

2) *Loss Settings*: The loss of PCDOL contains three parts: L_{con} to accommodate CD, L_{fs} to speed up computation, and L_{fr} to prevent CF. As shown in Table III, the absence of any one part is less effective than the whole under the ablation experiments. On task 0, the L_{con} and $L_{con} + L_{fs}$ approaches only outperform our approach by about 0.1%, and their continual

learning capability is significantly lower than that of PCDOL. In the final task, PCDOL can exceed a maximum accuracy of 36.72% and a minimum of 4.18%. Due to the feature replay item, PCDOL is effective in overcoming CF under the OCL process. When dealing with CD, PCDOL does not experience a dramatic decrease in accuracy compared to other methods. This shows that PCDOL is significantly better than others in terms of OCL ability.

3) *Online Continual Learning Capabilities*: The categories are gradually increased during the online continual learning process. The data stream is unbalanced. By comparing the accuracy and F1-score of each task as well as the final average accuracy and average F1-score, we find that our method is generally more effective than other methods. When the anomaly of drift occurs, as shown in Fig. 6, PCDOL is less affected than the other methods, and the overall loss decreases more quickly in trend. Fig. 2 compares the accuracy of each task on the ECG. Fig. 3 shows the F1-score for each task. Fig. 4 compares the accuracy of each task on PPG. We can find that the number of classes increases and the accuracy rate tends to decrease due to OCL. Compared with the continual learning method GEM, our accuracy is up to 22.4% higher. PCDOL is higher than EWC and RESERVOIR methods by 24.6% and 17.5%, respectively, and is up to 18.7% higher in average accuracy compared with the CoPE. Our average accuracy (dashed line) outperforms the other methods. The prototype suppression item allows models to not change drastically due to CD. Feature replay improves the model's OCL capability, and the accuracy of the model can still be higher than other methods even if the data increases. We can clearly find that GEM is affected by the size of the data, but PCDOL still performs well with unbalanced data.

As shown in Tables IV–VII, by comparing the average accuracy (Avacc), the average F1-score (AvF1), the average mae

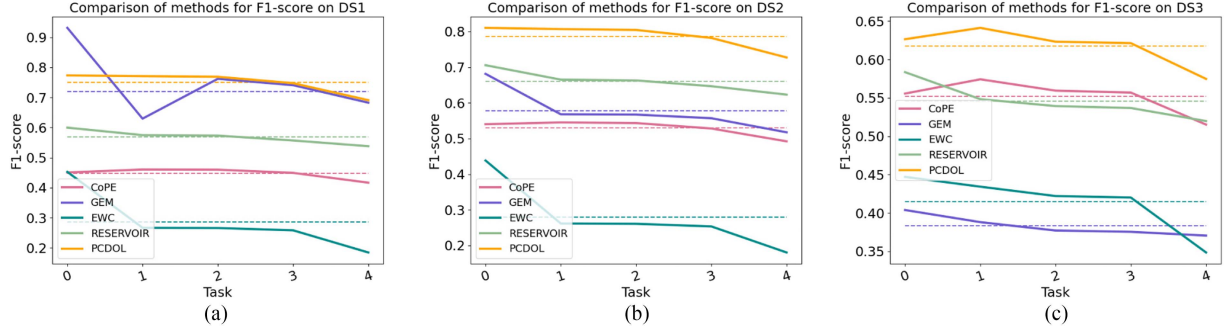


Fig. 3. F1-score for each task. (a) to (c) correspond to the F1-scores based on DS1 to DS3 of different methods.

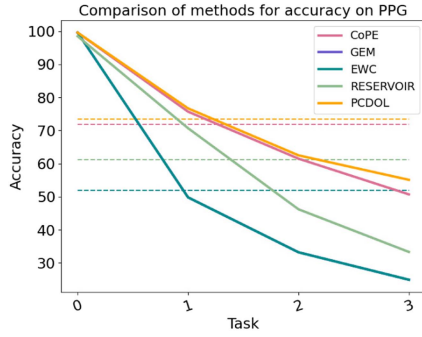


Fig. 4. Accuracy(%) for each task.

TABLE IV
EXPERIMENT DATA ON DS4

Method	Avacc(%)	AvF1	Avmae	Avmse
CoPE	50.4	0.3053	1.7718	5.1972
GEM	59.7	0.7812	0.6711	2.7003
EWC	40.1	0.4174	3.0679	21.132
RESERVOIR	58.7	0.6325	1.8614	11.657
PCDOL	69.1	0.6318	1.2213	4.7024

The bold values indicates the maximum values of the column.

TABLE V
EXPERIMENT DATA ON DS5

Method	Avacc(%)	AvF1	Avmae	Avmse
CoPE	55.3	0.4185	1.8824	7.6666
GEM	61.6	0.7778	0.8475	3.8478
EWC	43.4	0.2930	6.0615	15.602
RESERVOIR	48.2	0.6626	1.1163	4.7716
PCDOL	65.0	0.6449	1.1104	3.9450

The bold values indicates the maximum values of the column.

TABLE VI
EXPERIMENT DATA ON AR

Method	Avacc(%)	AvF1	Avmae	Avmse
CoPE	60.6	0.5727	0.9764	6.0037
GEM	57.2	0.6529	1.0536	4.6752
EWC	43.6	0.2536	3.8701	27.859
RESERVOIR	49.2	0.7256	1.0229	6.2722
PCDOL	66.7	0.8477	0.5120	2.6639

The bold values indicates the maximum values of the column.

TABLE VII
EXPERIMENT DATA ON PPG

Method	Avacc(%)	AvF1	Avmae	Avmse
CoPE	71.9	0.5449	0.6622	1.1390
GEM	51.9	0.4539	0.7849	1.3686
EWC	51.9	0.5208	1.4671	3.4559
RESERVOIR	61.2	0.3947	1.5356	6.3475
PCDOL	73.5	0.5453	0.6757	1.2031

The bold values indicates the maximum values of the column.

TABLE VIII
THE ROLE OF MEMORY

Method	memory size	accuracy(%)				Avacc(%)	time
		task0	task1	task2	task3		
PCDOL	100	99.6	73.4	63.7	49.7	71.6	41.982
	300	99.6	75.8	61.1	42.4	69.7	50.603
	500	99.6	69.4	52.5	42.8	66.1	49.902
CoPE	100	99.6	71.2	62.3	45.1	69.6	38.660
	300	99.6	71.6	64.1	48.5	71.0	50.781
	500	99.6	75.8	63.5	46.3	71.3	44.899

The bold values indicates the maximum values of the column.

(Avmae), and the average mse (Avmse), each data point of our task is overall higher than the comparison method.

4) *The Role of Prototype Suppression*: As shown in Fig. 5, when there is no prototype suppression item in the model (excl), the accuracy is significantly less than when it is included (incl). For the PPG real dataset, the accuracy starts to decrease due to the increase in data. In the final task, adding constraints improves 11.3% over no constraints. For ECG data, the accuracy without constraints is much lower than PCDOL for the real data set AR due to data imbalance. The accuracy without constraints is also lower than that of PCDOL under the influence of drift generated by the artificial dataset DS1. This verifies the necessity of setting (3). A bold value indicates the maximum value of the column. ■

5) *Time and Space Complexity Analysis*: Fig. 7 shows the data containing drift affects the model processing speed. PCDOL does not experience a dramatic increase in processing time due to the occurrence of CD. So PCDOL can still be energy efficient despite the inclusion of drifting data. We set the memory size according to the memory allocated to each class. The memory size will change according to the number of classes so that the memory space is used efficiently and adequately. In Table VIII,

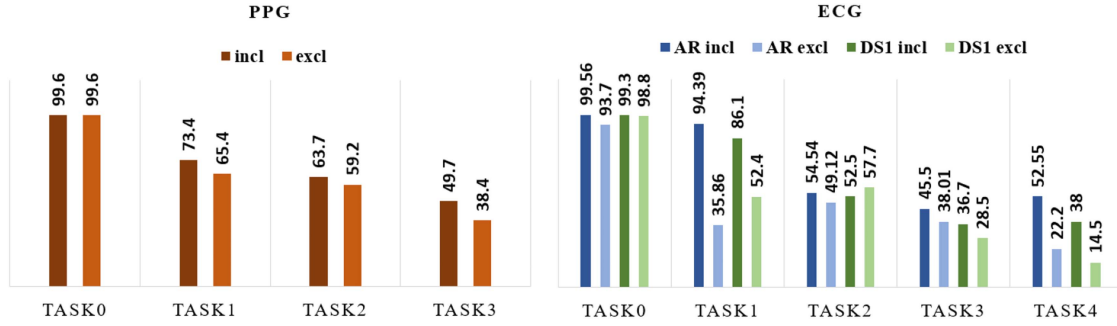


Fig. 5. Accuracy (%) for prototype suppression module included and not included.

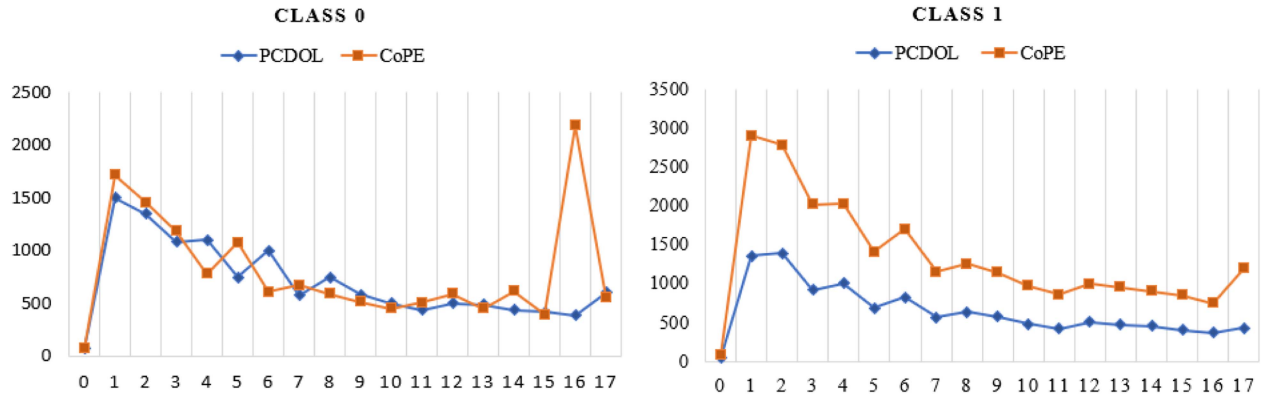


Fig. 6. Change in loss when drift occurs.

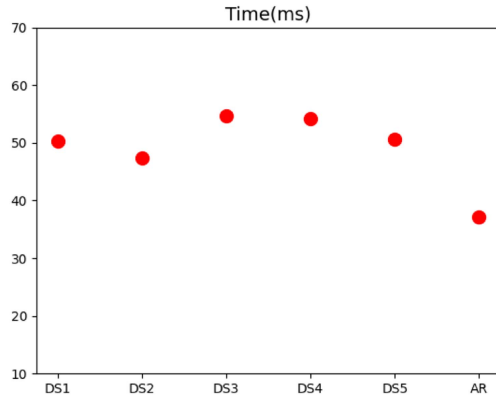


Fig. 7. Compare the model's processing speed for data with and without drift, where DS1-5 include drift and AR does not include drift.

VARIAION OF ACCURACY RATE(%) WITH DIFFERENT MEMORY

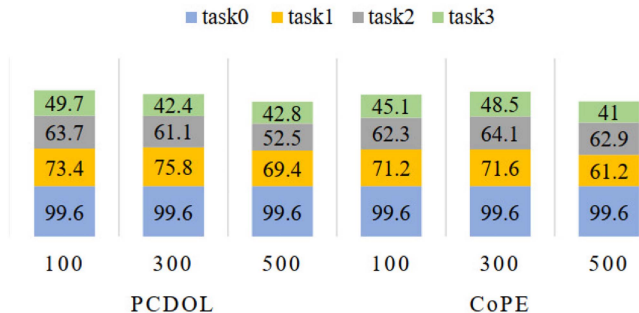


Fig. 8. Accuracy(%) for each task based on PCDOL and CoPE.

TABLE IX
DIFFERENT NUMBER OF TASKS PER TASK ON DS4

number	Avacc(%)	AvF1	Avmae	Avmse
three	55.3	0.5528	4.2376	11.2633
two	69.1	0.6318	1.2213	4.7024

The bold values indicates the maximum values of the column.

by comparing memory sizes of 100, 300, and 500 per class, PCDOL is well suited for the case of small memory. The trend of accuracy rate with different memory sizes is shown in Fig. 8. Unlike the CoPE method, PCDOL is less affected by the memory size variation generated, has a faster processing speed and consumes less memory.

6) Other Different Influencing Factors: We use EEG to test the classification effect of the algorithm on time series. The average accuracy can reach 64%, F1 is 0.7965, mae is 0.3919, and mse is 1.2781. As shown in Table IX, the increase in classes leads to a decrease in model learning ability due to the large disparity of classes in each task.

D. Case Study

PCDOL can be deployed on nanorobots to classify medical time series in real-time. Real-time data collection is achieved by embedding the nanorobots in the human body. The nanorobots move through the blood vessels to detect blood glucose levels and provide timely warnings to prevent diabetes [1]. Nanorobots embedded with biosensors can detect early tumor cells by collecting the intensity of E-cadherin signals [33]. The nanomedical heart tracker collects ECG and analyzes it through nanoelectrodes [34]. The acquisition of ECG and PPG data by

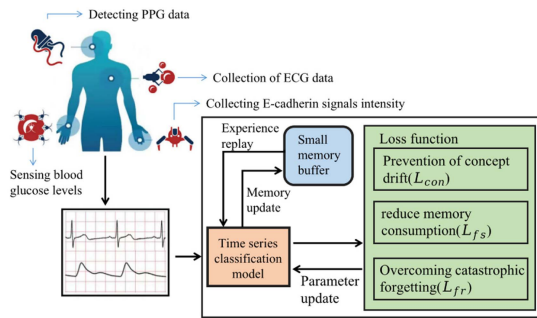


Fig. 9. PCDOL workflow on nanorobots.

the nanorobot allows continuous estimation of blood pressure based on pulse arrival time without the need for additional equipment [35]. Nanorobots enable remote health testing. As shown in Fig. 9, the monitoring, analysis, and early warning of physiological signals can be realized in various scenarios by the real-time analysis function that comes with the algorithm.

V. CONCLUSION

To achieve real-time classification of dynamically changing time-series signals on nanorobots, we propose an OCL algorithm, PCDOL. Through the prototype design based on clustering, we keep the prototype features and memory updated to realize real-time online processing. PCDOL can adapt to the potential CD in the data. During OCL, it can effectively handle catastrophic forgetting. Also, it is energy efficient and lightweight, making it suitable for application on nanorobots. In the future, we will improve the classification performance of PCDOL on tasks containing different numbers of classes, improve the ability of PCDOL to accurately predict the occurrence of concept drift, and replace the model.

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